

SOURCES OF ERROR

1. Graduation marks may not have equal separation.
2. The lines showing graduations may not be sharp as required.

ACTIVITY 2

AIM

To determine the mass of a given body using a metre scale by principle of moments.

YOU NEED

1. A body of unknown mass
2. A metre scale
3. A broad heavy wedge with sharp edge.

THEORY

Let m is the mass of the given body. Let a metre rod be supported on a sharp wedge at its centre of gravity G . If unknown mass m of weight $W = mg$ is suspended in the left side of the metre rod and a known standard mass M of weight $W_1 = Mg$ is suspended in the right side of the metre rod in such a way so that the metre rod is balanced, i.e., it remains horizontal as shown in Fig. 46.

Now by applying the principle of moment we have

$$W \times a = W_1 \times b \quad \text{or} \quad (mg) \times a = (Mg) \times b$$

$$\text{or} \quad m = \frac{(Mg) \times b}{ga} \quad \text{or} \quad m = M \left(\frac{b}{a} \right)$$

where a is the distance of unknown mass from centre of gravity and b is the distance of known mass from the centre of gravity.

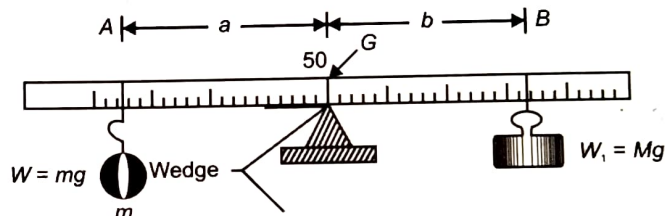


Fig. 46. Determination of mass using metre scale

HOW TO DO

1. Put the sharp wedge on the wooden block so that sufficient space remains below the scale for hanging weights and body of unknown mass.
2. Now place the scale on the wedge with its graduated face up.
3. Now adjust the position of the scale so that the scale remains balanced, i.e., it remains in horizontal position, when no weights are suspended. In this position the centre of gravity of the scale is just above the wedge, this is called the equilibrium position of the scale. Record the position of the wedge on the scale.
4. This reading gives the position of centre of gravity G of the metre scale. During the experiment the metre scale is always placed on this position of the centre of gravity G on the wedge.
5. Now suspend the unknown weight $W = mg$ on the left side of the wedge and a known standard weight $W_1 = Mg$ on the right side of the wedge, with the help of the cotton thread, as shown in Fig. 46.
6. Now adjust the position of unknown mass m so that the scale gets balanced in the horizontal position.
7. When the horizontal position of the metre scale becomes stable, record the position of body of the unknown mass A , known mass B and centre of gravity of the scale G on the metre scale. Note $AG = a$ and $GB = b$.
8. Now repeat the steps 3, 4, 5, 6 and 7 by changing the positions of unknown weight $W = mg$ and known weight $W_1 = Mg$. Record the observations in a tabular form in the observation table.

OBSERVATIONS

1. Position of the centre of gravity of the scale = $G = \dots \text{ cm}$.

2. Value of the standard mass M used = g.

$$W_1 = Mg = \dots \text{ gf.}$$

Table for unknown mass m

S.No.	Position of the body of unknown mass ' m ' A (cm)	Position of known mass M B (cm)	$a = AG$ (cm)	$b = GB$ (cm)	$m = \left(\frac{b}{a}\right) M$ (g)
1.					
2.					
3.					
4.					

CALCULATIONS

Mean mass

$$m = \frac{m_1 + m_2 + m_3 + m_4}{4} = \dots \text{ g}$$

RESULT

The value of the unknown mass m of the given body by using metre scale is g.

BE CAREFUL

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1. The edge of the wedge should be perpendicular to the graduation scale.
2. The wedge used for balancing the scale should have sharp edge and it should be perpendicular to the length of metre scale.
3. The reading should be taken only when the metre scale becomes stable and in the horizontal position.
4. The known and unknown weights should not be so heavy as to bend the scale.
5. The centre of gravity of the scale should be determined accurately and the scale should be placed at this position on the wedge.
6. Some part of the cotton thread over the scale should be kept along the graduations of the scale.

SOURCES OF ERROR

1. The edge may not be sharp.
2. Metre scale may have faulty calibration.
3. The thread used may be thick and heavy.